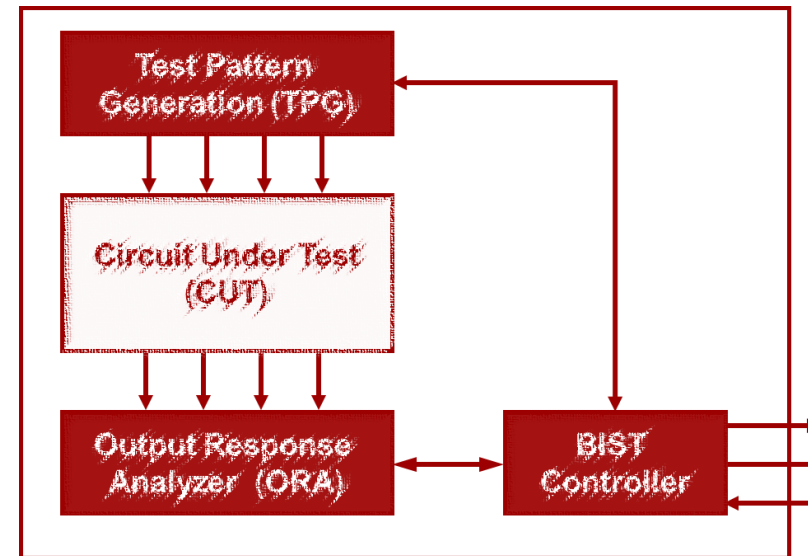


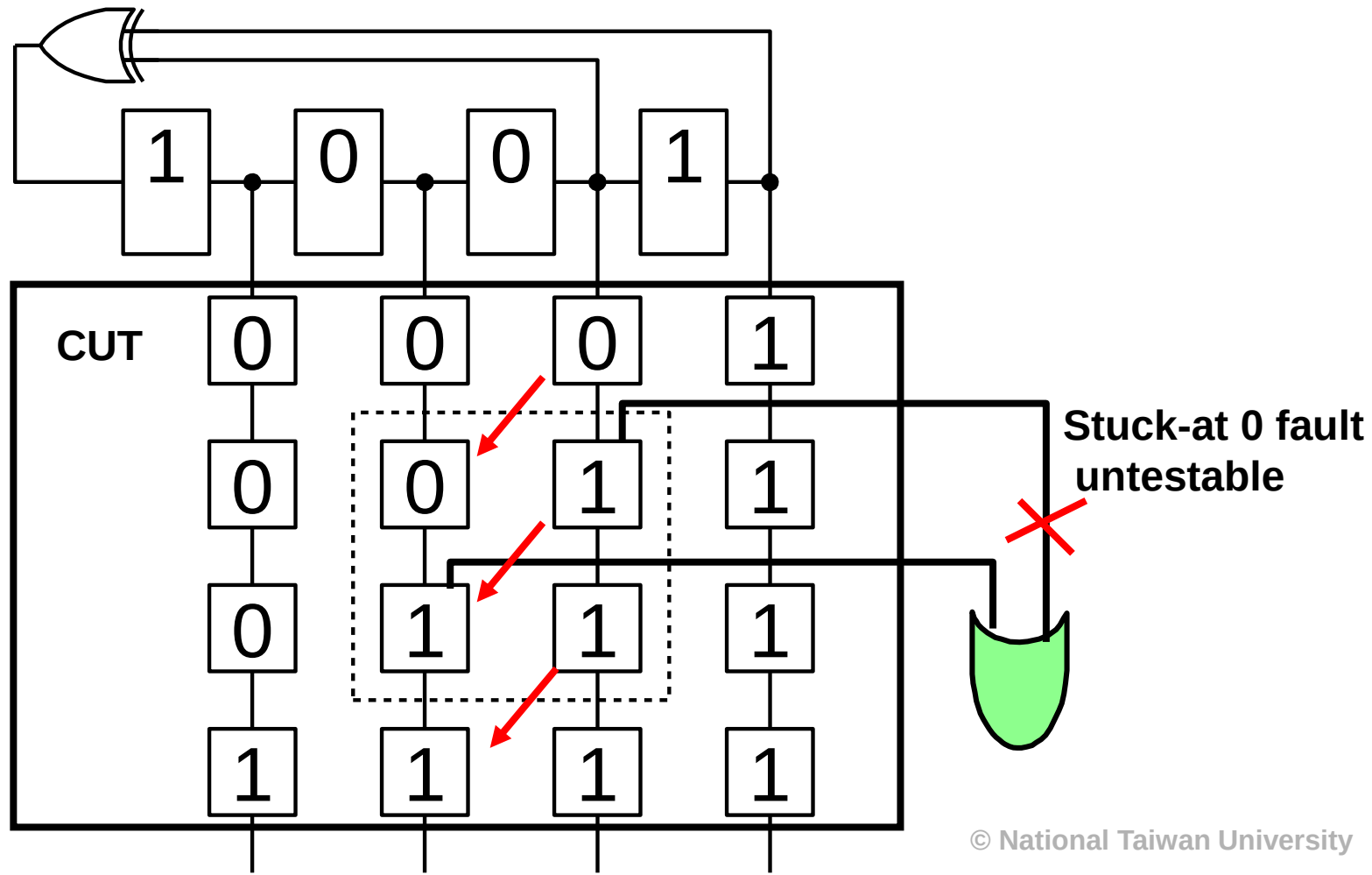
BIST Part 2

- Introduction
- Pattern Generation
- Output Response Analysis
- BIST Architecture
- **Problems and Solutions**
 - ◆ Fault coverage not high enough
 - * Structure Dependency
 - * Linear Dependency
 - * Random Pattern Resistant Fault
 - ◆ “X” Problem
 - ◆ Area overhead
 - ◆ Long Test length
 - ◆ Diagnosis/Debug
- Conclusions



Structure Dependency Problem

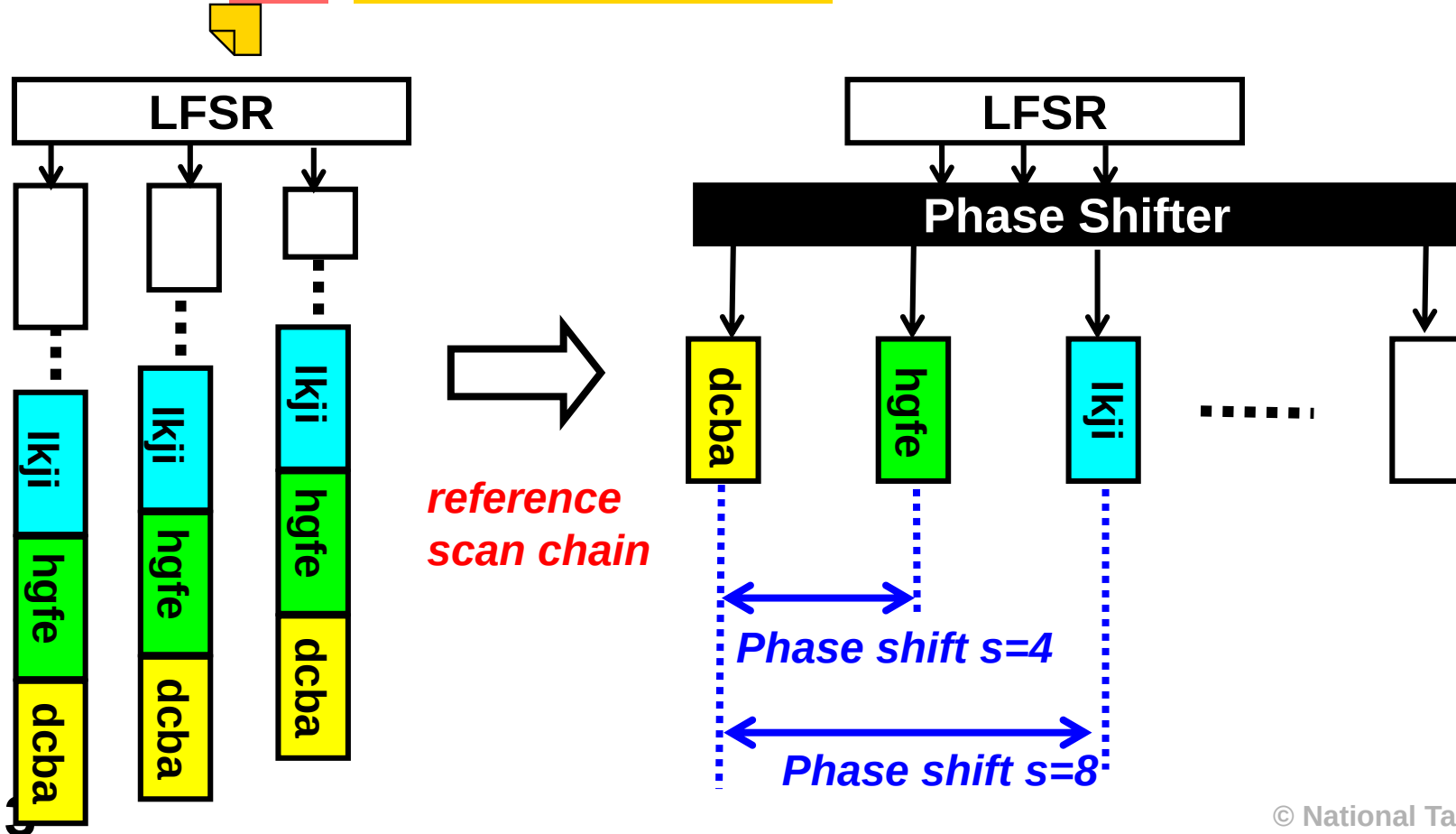
- A chain is shifted version of another chain
 - ♦ Some faults become untestable due to structure dependency



Solution: *Phase Shifter* [Bardell 87]

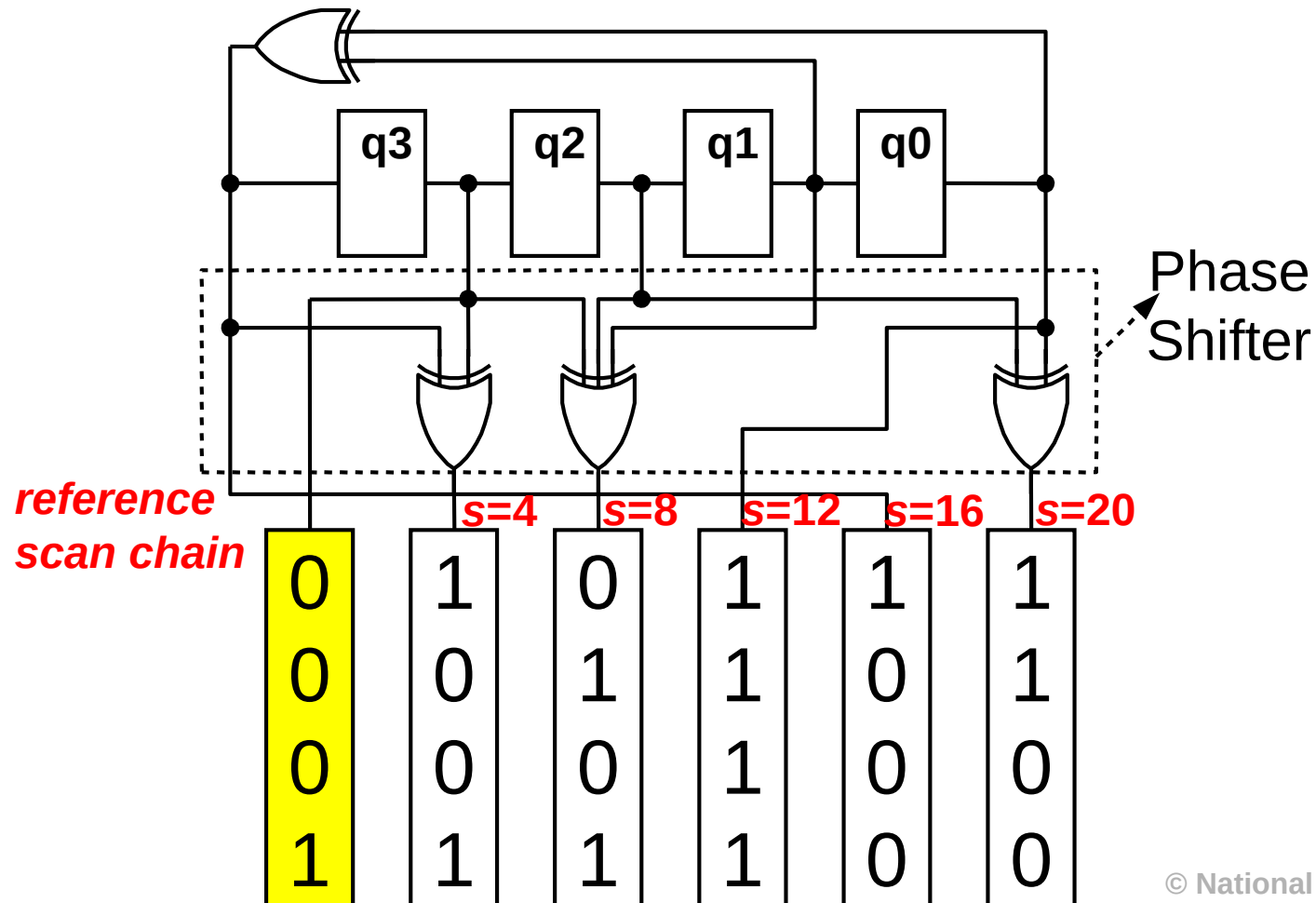
- Modified *STUMPS* : insert phase shifter between LFSR and CUT
- Phase shifter is a *serial to parallel converter*
 - ♦ Each scan chain is phase shifted **s** cycles

* w.r.t. *reference scan chain*



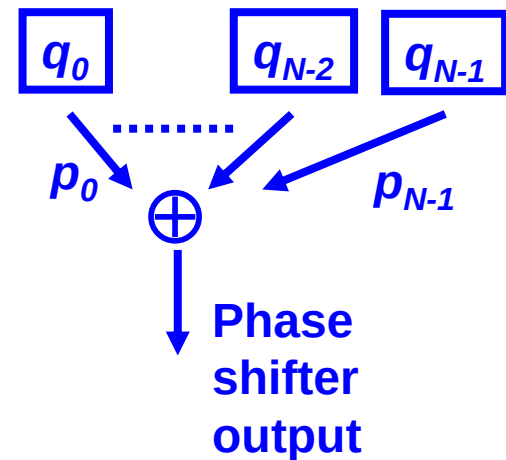
What Is Inside a Phase Shifter?

- **PS** is simply implemented by XOR net work
- Why? M-sequence property:
 - ♦ $\text{M-sequence}_{\text{PS1}} + \text{M-sequence}_{\text{PS2}} = \text{M-sequence}_{\text{PS3}}$



Design a Phase Shifter

- Given an N -degree LFSR , T is companion matrix
 - $Q = [q_0 \ q_1 \ q_2 \ \dots \ q_{N-1}]^T$, Q is a column vector of FF states
 - LFSR after s cycles $T^s Q$
- Given a reference scan chain Only one element in B is 1.
 - B is **selection vector**, $B = [b_0 \ b_1 \ b_2 \ \dots \ b_{N-1}]$
 - $b_i = 1$ means q_i is reference scan chain
 - Reference scan chain output = BQ
- Phase shifter row vector $P = [p_0 \ p_1 \ p_2 \ \dots \ p_{N-1}]$
 - $p_i = 1$ is a tap point from flip-flop q_i
 - Phase shifter output = PQ Multiple elements in P can be one.
- PS output shifted by s cycles w.r.t. reference scan chain
 - So, $PQ = BT^s Q$



$$\text{Phase Shifter : } P = BT^s$$

Example

- Given LFSR $1+x+x^4$
 - Reference scan output from q_3

- $N=4, s=4, T=$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}$$

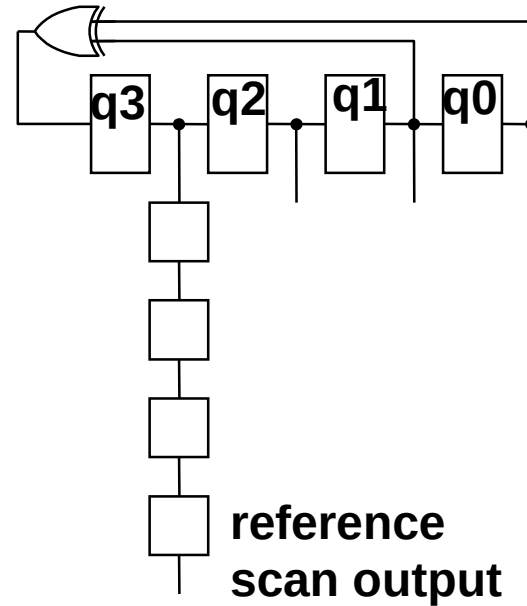
- $B = [0 \ 0 \ 0 \ 1]$

- $P_{s=4} = BT^4 = [0 \ 0 \ 0 \ 1]$

$$\begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \end{bmatrix}^4$$

$$= [0 \ 0 \ 0 \ 1] \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}^2$$

$$= [1 \ 1 \ 0 \ 1] \quad \text{tap points } q_0, q_1, q_3$$

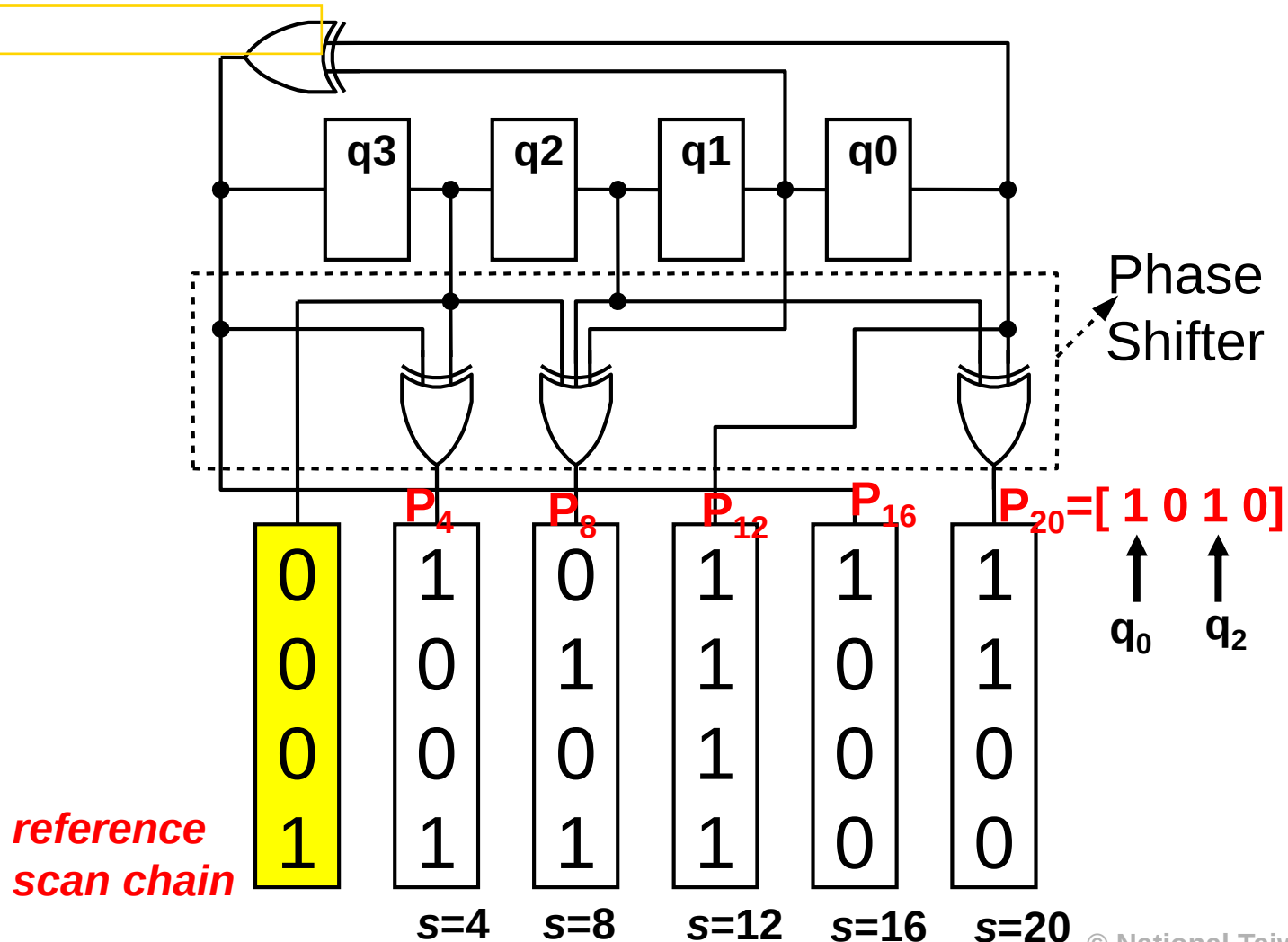


6 Similarly, $P_{s=8} = BT^4T^4 = [0 \ 1 \ 1 \ 1]$

Example (cont'd)

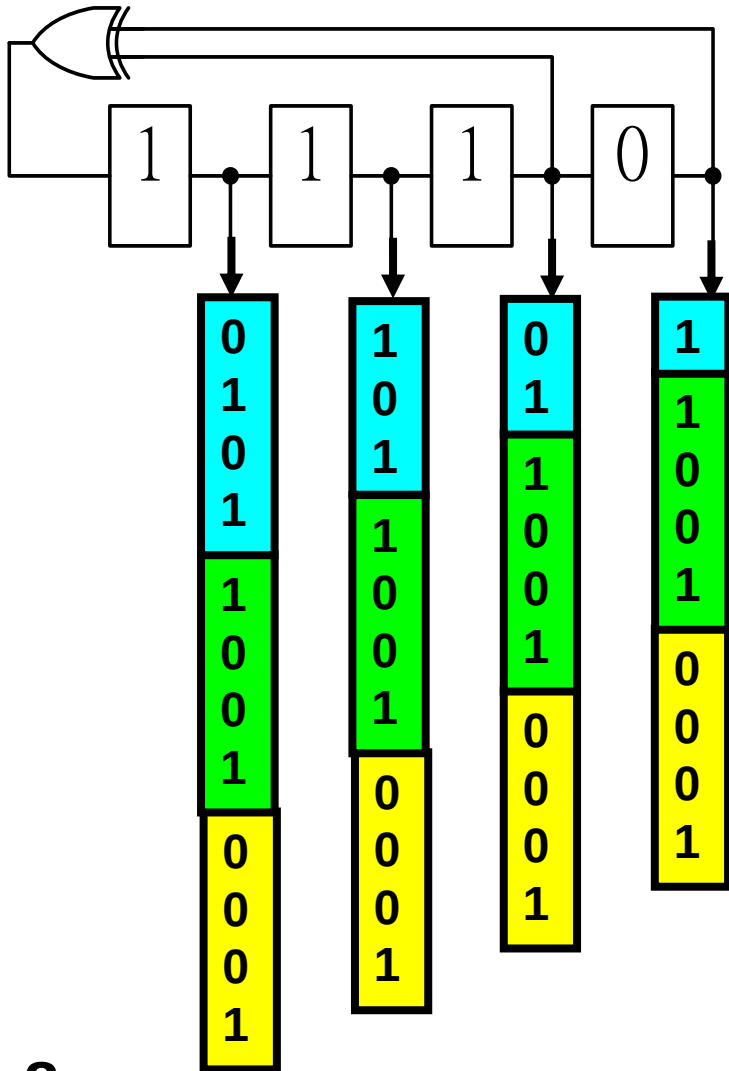
$$P_{s=4} = [1101] \quad P_{s=8} = BT^8 = [0111] \quad P_{s=12} = BT^{12} = [1000]$$

$$P_{s=16} = BT^{16} = [1100] \quad P_{s=20} = BT^{20} = [1010]$$

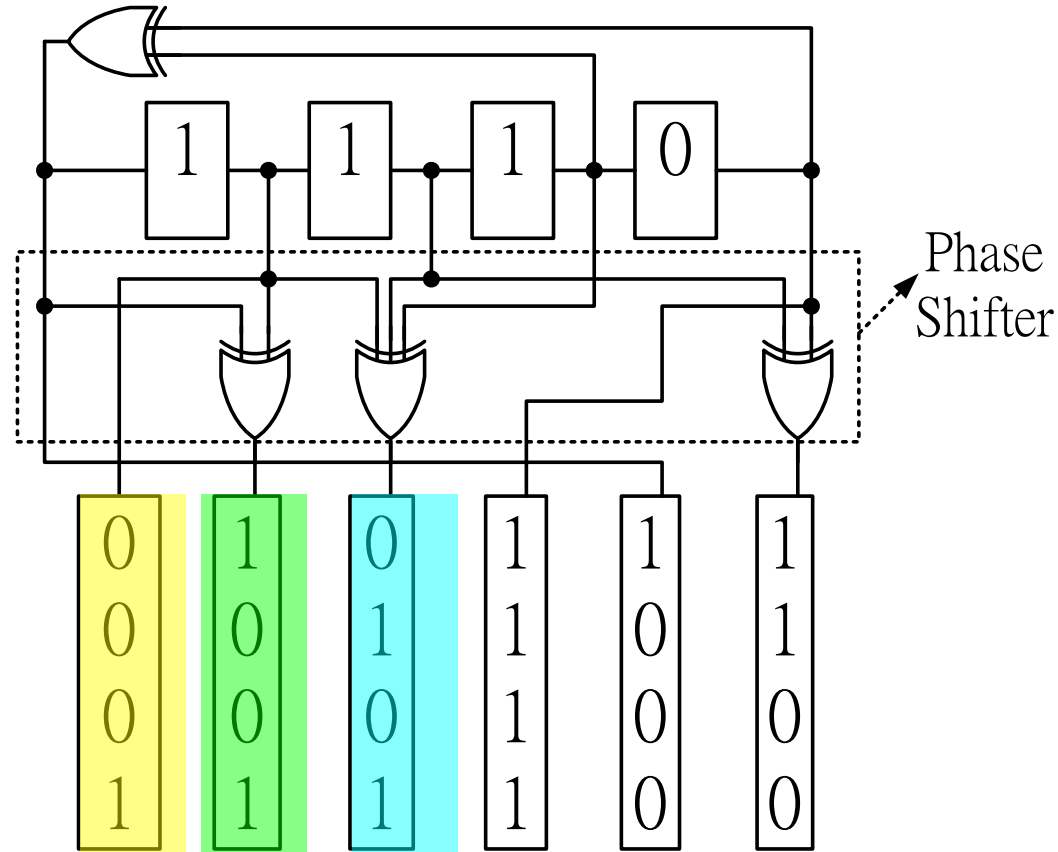


Comparison

- Without Phase Shifter



- With Phase Shifter



Phase Shifter Summary

- Phase Shifter benefits
 - ◆ Reduce structure dependency, increase randomness
 - ◆ Small area overhead
 - ◆ Support more scan chains than LFSR degree N
- Some research claims **LFSR + phase shifter**
 - ◆ provides similar randomness as **CA**

PS Increases Randomness at Small cost